

CyberGuardian AI

Intelligent Threat Detection and Automated Incident Response Platform

Problem Statement Title	--	AI-Powered Behavioral Anomaly Detection for Cybersecurity
Theme	--	Artificial Intelligence & Machine Learning (AI/ML) , Cybersecurity
PS Category	--	Software
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AI-Powered Behavioral Anomaly Detection for Cybersecurity

User & Network Data



AI/ML Behavioral Analysis



Anomaly Detection



Risk Scoring



MITRE ATT&CK Mapping



Alerts & Incident Report

Proposed Solution

An AI-powered platform that continuously monitors user and network activities, learns normal behavior patterns, detects anomalies in real time, assigns risk scores, maps threats to the MITRE ATT&CK framework, and automatically generates incident reports for rapid response.

How It Addresses the Problem

Detects unknown, zero-day, and insider threats using behavioral analysis.
Reduces response time through continuous real-time monitoring.
Minimizes false positives using adaptive machine learning.
Enables proactive threat detection before attacks escalate.

Innovation & Uniqueness

AI-driven behavioral analytics instead of signature-based detection.
Self-learning models that adapt to evolving cyber threats.
Integrated MITRE ATT&CK mapping with automated incident reporting.
Scalable solution for enterprise, cloud, and hybrid environments.

TECHNICAL APPROACH

Technologies Used

- Programming Language: Python
- AI/ML: Scikit-learn, Pandas, NumPy
- Visualization: Plotly, Matplotlib
- Backend: Flask
- Threat Intelligence: MITRE ATT&CK Framework
- Data Processing: JSON, CSV Log Analysis
- Development Tools: VS Code, Git, GitHub

Implementation Methodology

Security Logs & Network Events



Data Preprocessing & Feature Extraction



AI Behavioral Analysis Engine



Threat Prediction & Risk Scoring



MITRE ATT&CK Technique Mapping



Automated Incident Report Generation



Live Security Dashboard & Alerts

FEASIBILITY AND VIABILITY

Feasibility

- Built using proven technologies: Python, Flask, Scikit-learn, MITRE ATT&CK.
- Easily integrates with existing enterprise security infrastructure.
- Supports cloud, on-premise, and hybrid deployment.
- Scalable architecture for organizations of different sizes.

Challenges & Risks

- Limited availability of quality cybersecurity training data.
- False positives during early model deployment.
- Rapidly evolving cyberattack techniques.
- High-volume real-time log processing.

Mitigation Strategies

- Continuously retrain AI models with updated threat intelligence.
- Use analyst feedback to improve detection accuracy.
- Update MITRE ATT&CK mappings to address new attack patterns.
- Optimize the detection pipeline for fast, scalable performance

Solution Workflow

Feasible Technology



AI Threat Detection



Continuous Learning



Enterprise Deployment

ARTIFACTS

```
(.venv) PS C:\Users\nayar\OneDrive\Documents\HONEYWELL> python generator/live_stream.py
CyberGuardian AI Real-Time Threat Stream Started

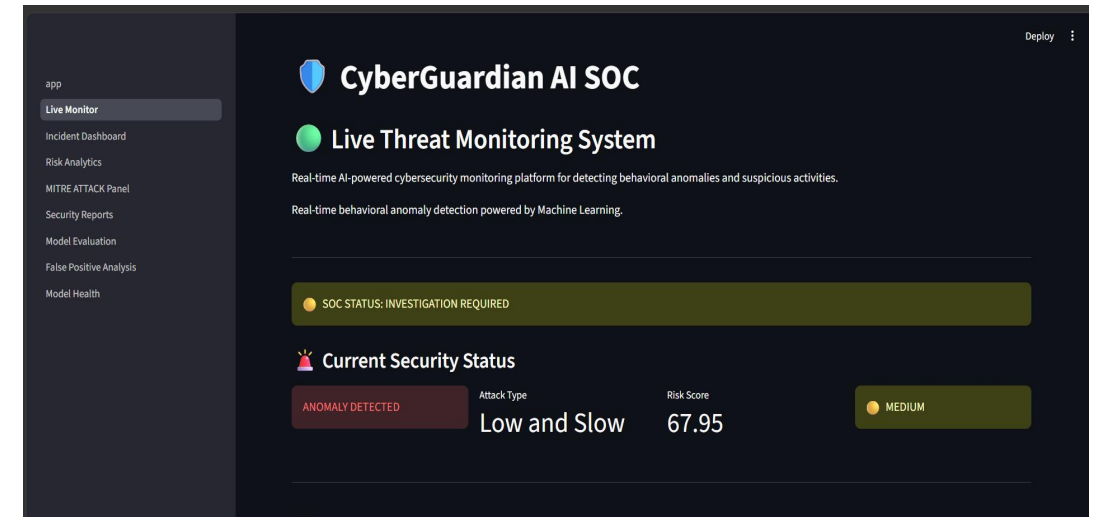
-----
Selected Event:
95614  attack_type
Insider Drift
-----
Status: NORMAL
Attack: Normal
Risk: 13.64
Threat: LOW
-----

Selected Event:
66571  attack_type
Unknown
-----
Status: NORMAL
Attack: Normal
Risk: 14.04
Threat: LOW
-----

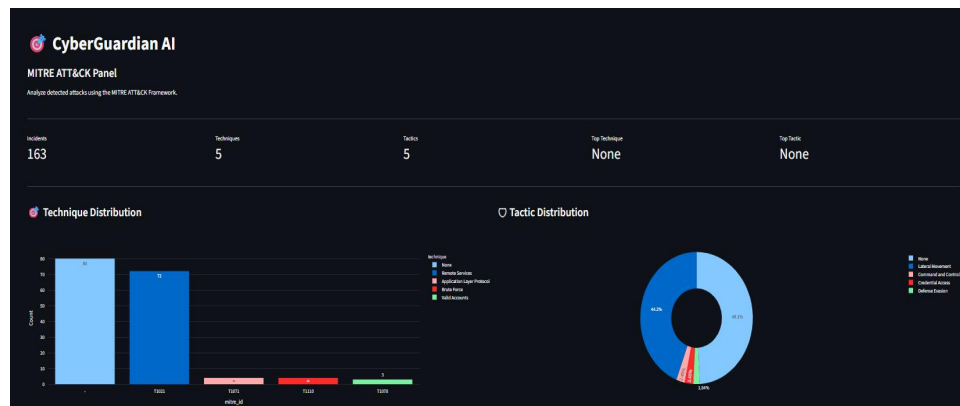
Selected Event:
97471  attack_type
Low and Slow
-----
Status: NORMAL
Attack: Normal
Risk: 25.75
Threat: LOW
-----

Selected Event:
22009  attack_type
Unknown
-----
Status: NORMAL
Attack: Normal
Risk: 15.54
Threat: LOW
-----
```

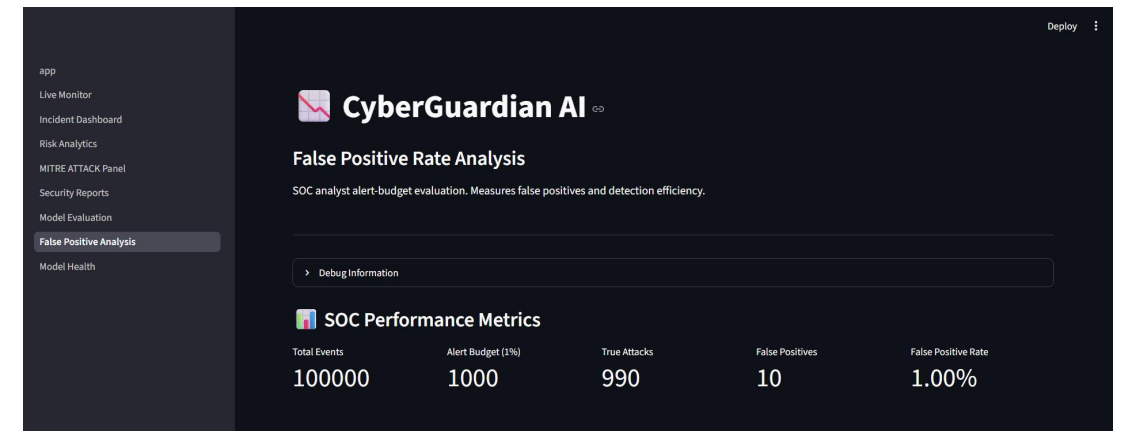
1. AI Threat Detection Output



2. Live Security Dashboard



3. MITRE ATT&CK Mapping



4. False Positive Rate Analysis

RESEARCH AND REFERENCES

Research Sources

MITRE ATT&CK Framework – <https://attack.mitre.org>

NIST Cybersecurity Framework (CSF) – <https://www.nist.gov/cyberframework>

OWASP Foundation – <https://owasp.org>

Scikit-learn Documentation – <https://scikit-learn.org>

Python Documentation – <https://docs.python.org/3>

Research Papers

1. [https://www.researchgate.net/profile/Abdullahi-Abdulkareem-2/publication/394584633_AI-Powered Cybersecurity in Edge Computing Lightweight Neural Models for Anomaly Detection/links/68c0c2976f87c42f3b914e77/AI-Powered-Cybersecurity-in-Edge-Computing-Lightweight-Neural-Models-for-Anomaly-Detection.pdf](https://www.researchgate.net/profile/Abdullahi-Abdulkareem-2/publication/394584633_AI-Powered_Cybersecurity_in_Edge_Computing_Lightweight_Neural_Models_for_Anomaly_Detection/links/68c0c2976f87c42f3b914e77/AI-Powered-Cybersecurity-in-Edge-Computing-Lightweight-Neural-Models-for-Anomaly-Detection.pdf)
2. [https://www.researchgate.net/profile/Anjani-Kumar-Polinati/publication/391701080_AI-Powered Anomaly Detection in Cybersecurity Leveraging Deep Learning for Intrusion Prevention/links/6837bada6b5a287c30473630/AI-Powered-Anomaly-Detection-in-Cybersecurity-Leveraging-Deep-Learning-for-Intrusion-Prevention.pdf](https://www.researchgate.net/profile/Anjani-Kumar-Polinati/publication/391701080_AI-Powered_Anomaly_Detection_in_Cybersecurity_Leveraging_Deep_Learning_for_Intrusion_Prevention/links/6837bada6b5a287c30473630/AI-Powered-Anomaly-Detection-in-Cybersecurity-Leveraging-Deep-Learning-for-Intrusion-Prevention.pdf)